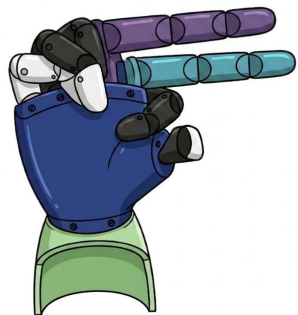




BuildBuddy @ Dartmouth

*AI-Powered ASL Fingerspelling Hand
Building Instructions
Designed for Dartmouth Students!*





H



E



L



L

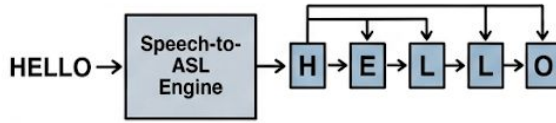


O

1. INPUT:



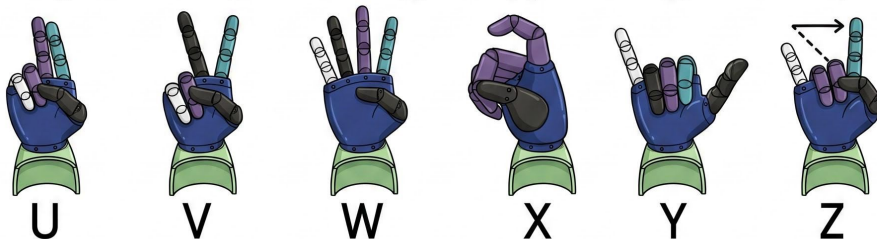
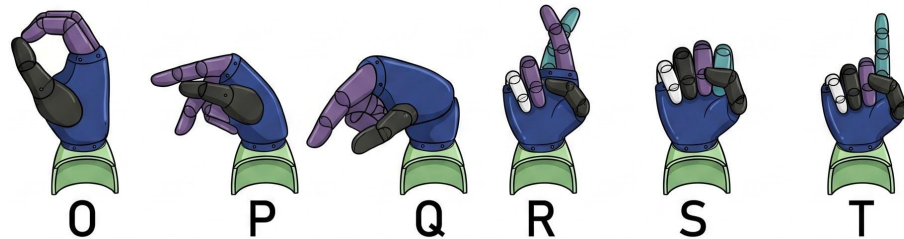
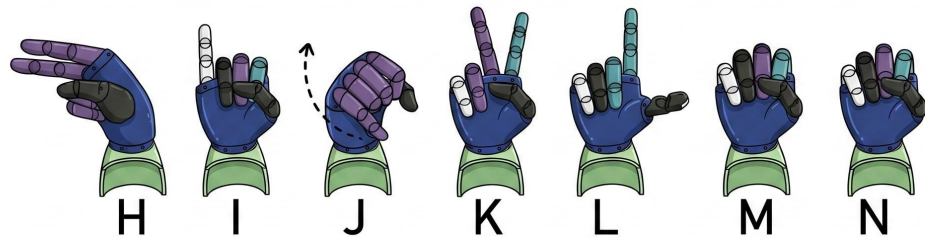
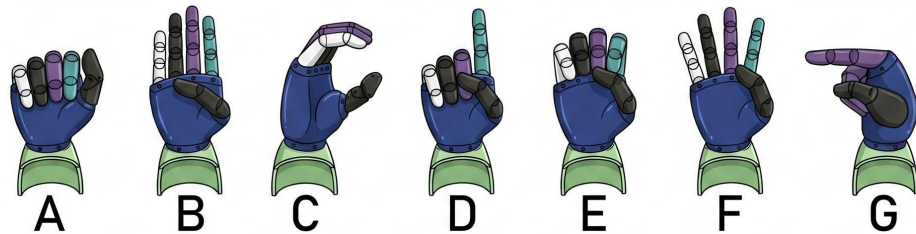
2. INTERPRETATION:



3. OUTPUT (SIGNING):



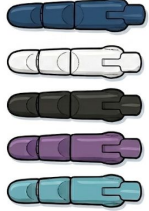
Build a 3D-printed hand that turns voice input into the ASL alphabet *in 24 hours!*



**Learn the ASL
alphabet!
Learn to build!**



Parts Needed



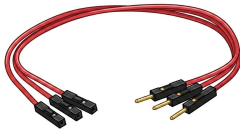
3D-printed hand parts
(see next page)



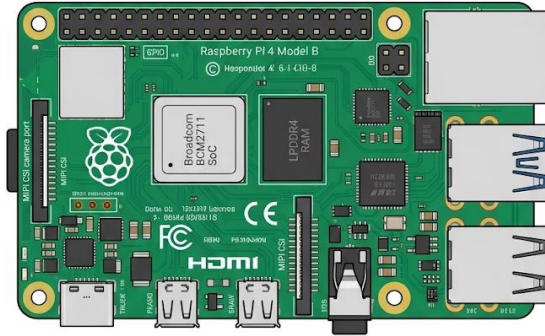
Bungee Cord



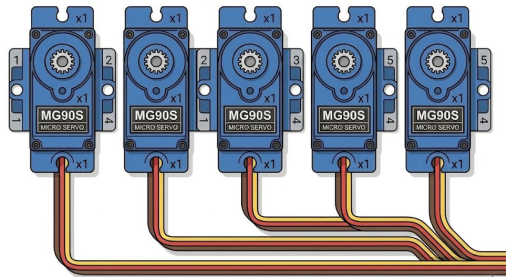
Fishing Line



Male-to-Female
Dupont Jumper
Wires



Raspberry Pi 4 Model B



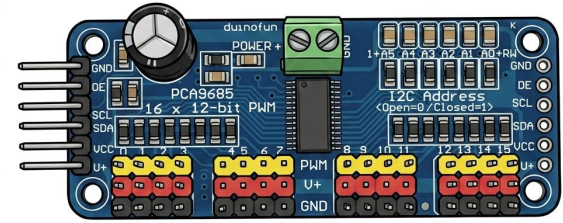
5x MG90S Micro Servos



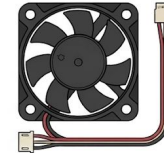
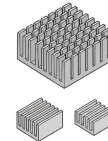
5V 5A Power Supply
with Barrel Jack Adapter



USB
Microphone



PCA9685



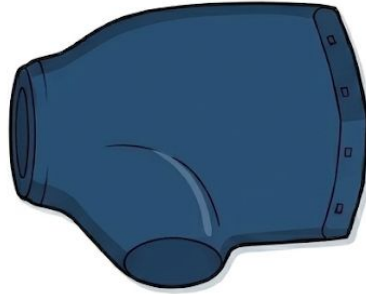
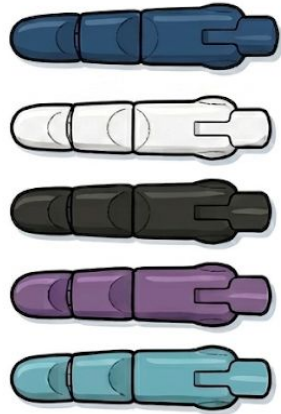
Heatsinks and Fan
for Pi



USB-C Power
Supply for Pi



3D-Printed Hand Parts



Download and
print parts from

<https://tinyurl.com/2v6u7m3e>



3D-Printing @ Dartmouth

Go to MShop (C027) or Harold Edward Cable MakerSpace (ECSC 003), give the STL files from the previous page, and get them printed!



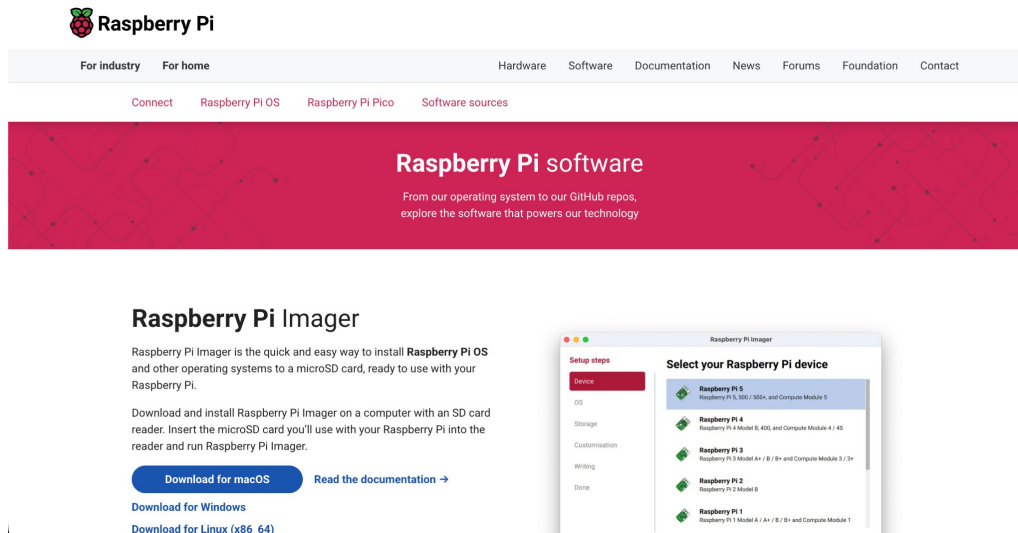
Part



Set Up the Raspberry Pi

Download Raspberry Pi Imager from raspberrypi.com/software

1



The image shows the Raspberry Pi website's software section and the Raspberry Pi Imager application. The website has a red header with the Raspberry Pi logo and navigation links. Below the header is a red banner with the text "Raspberry Pi software" and a sub-header "From our operating system to our GitHub repos, explore the software that powers our technology". The main content area is white and features the "Raspberry Pi Imager" section. This section includes a paragraph describing the imager, a list of download links for macOS, Windows, and Linux, and a link to read the documentation. To the right of the text is a screenshot of the Raspberry Pi Imager application window, which shows a list of Raspberry Pi devices to select from.

Raspberry Pi

For industry For home Hardware Software Documentation News Forums Foundation Contact

Connect Raspberry Pi OS Raspberry Pi Pico Software sources

Raspberry Pi software

From our operating system to our GitHub repos, explore the software that powers our technology

Raspberry Pi Imager

Raspberry Pi Imager is the quick and easy way to install **Raspberry Pi OS** and other operating systems to a microSD card, ready to use with your Raspberry Pi.

Download and install Raspberry Pi Imager on a computer with an SD card reader. Insert the microSD card you'll use with your Raspberry Pi into the reader and run Raspberry Pi Imager.

[Download for macOS](#) [Read the documentation →](#)

[Download for Windows](#)

[Download for Linux \(x86_64\)](#)

Raspberry Pi Imager

Setup steps

- Device
- OS
- Storage
- Customisation
- Writing
- Done

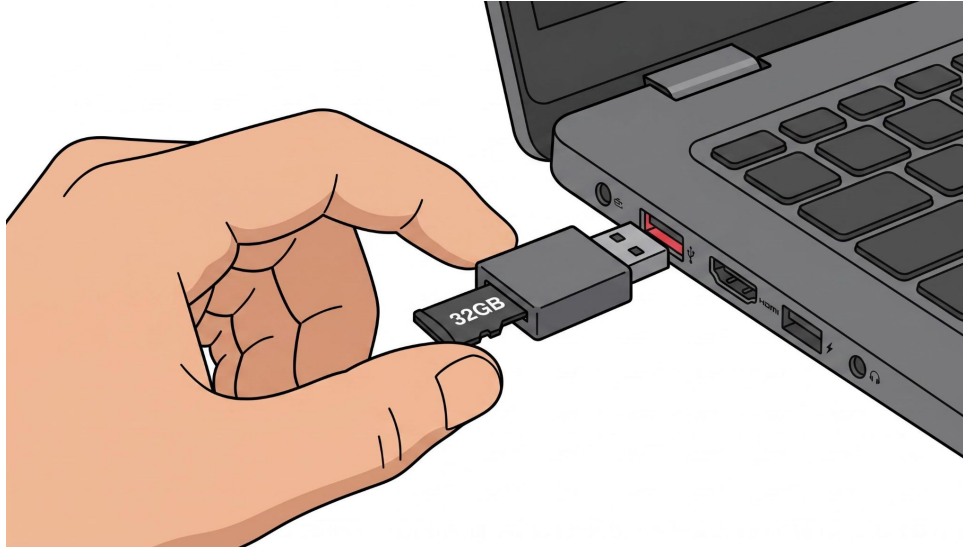
Select your Raspberry Pi device

- Raspberry Pi 5**
Raspberry Pi 5, 5B+, 5B+, and Compute Module 5
- Raspberry Pi 4**
Raspberry Pi 4 Model B, 4B, and Compute Module 4 / 4B
- Raspberry Pi 3**
Raspberry Pi 3 Model A+ / B+ and Compute Module 3 / 3+
- Raspberry Pi 2**
Raspberry Pi 2 Model B
- Raspberry Pi 1**
Raspberry Pi 1 Model A / A+ / B+ and Compute Module 1



Plug in your 32 GB micro SD card into your laptop
(you may need a micro SD to SD card adapter)

2



Open Raspberry Pi Imager

Choose Device -> Raspberry Pi 4

3

Setup steps

Device

OS

Storage

Customisation

Writing

Done

APP OPTIONS

Select your Raspberry Pi device



Raspberry Pi 5

Raspberry Pi 5, 500 / 500+, and Compute Module 5



Raspberry Pi 4

Raspberry Pi 4 Model B, 400, and Compute Module 4 / 4S



Raspberry Pi 3

Raspberry Pi 3 Model A+ / B / B+ and Compute Module 3 / 3+



Raspberry Pi 2

Raspberry Pi 2 Model B



Raspberry Pi 1

Raspberry Pi 1 Model A / A+ / B / B+ and Compute Module 1

NEXT



Open Raspberry Pi Imager

Choose OS -> Raspberry Pi OS (64-bit)

4

Setup steps

Device

OS

Storage

Customisation

Writing

Done

APP OPTIONS

Choose operating system

Select an operating system to install on your Raspberry Pi



Raspberry Pi OS (64-bit)

A port of Debian Trixie with the Raspberry Pi Desktop (Recommended)

Cached on your computer

Released: 2026-04-21



Raspberry Pi OS (32-bit)

A port of Debian Trixie with the Raspberry Pi Desktop

Online - 1.2 GB download

Released: 2026-04-21



Raspberry Pi OS (Legacy, 32-bit)

A port of Debian Bookworm with security updates and desktop environment

Online - 1.2 GB download

Released: 2026-04-13



Raspberry Pi OS (other)

Other Raspberry Pi OS based images

BACK

NEXT



Open Raspberry Pi Imager
Choose Storage -> {your micro
SD card}

5

Setup steps

Device

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

APP OPTIONS

Select your storage device



Apple SDXC Reader Media

28.2 GB

Mounted as /Volumes/bootfs



Exclude system drives

BACK

NEXT



Open Raspberry Pi Imager
Customisation → hostname →
“aslhand”

6

Device

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

APP OPTIONS

Customisation: Choose hostname

Enter your hostname

aslhand

A hostname is a unique name that identifies your Raspberry Pi on the network. It should contain only letters, numbers, and hyphens.

SKIP CUSTOMISATION

BACK

NEXT



Open Raspberry Pi Imager

Customisation → Localisation

7

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

APP OPTIONS

Customisation: Localisation

Select your location for suggested time zone and keyboard layout

Capital city:

Washington, D.C. (United States) ▼



Time zone:

America/New_York ▼

Keyboard layout:

us ▼

SKIP CUSTOMISATION

BACK

NEXT



Open Raspberry Pi Imager

Customisation → User →
Choose username

8

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

APP OPTIONS

Customisation: Choose username

Create a user account for your Raspberry Pi

Username:

Enter your username

ashleypi

Password:

Saved (hidden) – leave blank to keep

Confirm password:

Re-enter to change password

The username must be lowercase and contain only letters, numbers, underscores, and hyphens.

SKIP CUSTOMISATION

BACK

NEXT



Open Raspberry Pi Imager

Customisation → Wi-Fi →
Enter your Wi-Fi

9

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

APP OPTIONS

Customisation: Choose Wi-Fi

SECURE NETWORK

OPEN NETWORK

SSID:

Network name

KCC7

Password:

Saved (hidden) – leave blank to keep

Confirm password:

Re-enter to change password

☐ Hidden SSID

SKIP CUSTOMISATION

BACK

NEXT



Open Raspberry Pi Imager

Customisation → SSH

authentication → Use password
authentication

10

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

APP OPTIONS

Customisation: SSH authentication

Configure SSH access

Enable SSH

[Learn about SSH](#)



Authentication mechanism:

☒ Use password
authentication

☐ Use public key
authentication

SKIP CUSTOMISATION

BACK

NEXT



Open Raspberry Pi Imager

Customisation → Raspberry Pi
Connect → Off

11

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

[Raspberry Pi Connect](#)

Writing

Done

APP OPTIONS

Customisation: Raspberry Pi Connect

Sign in to receive a token and enable Raspberry Pi Connect

Enable Raspberry Pi Connect

What is Raspberry Pi Connect?



SKIP CUSTOMISATION

BACK

NEXT



Open Raspberry Pi Imager

Writing → Write

12

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

APP OPTIONS

Write image

Review your choices and write the image to the storage device

Summary

Device: **Raspberry Pi 4**
Operating system: **Raspberry Pi OS (64-bit)**
Storage: **Apple SDXC Reader Media**

Customisations to apply:

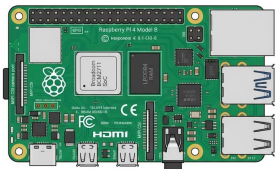
- Hostname configured
- Localisation configured
- User account configured
- Wi-Fi configured
- SSH enabled

BACK

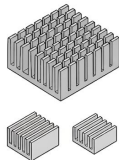
WRITE



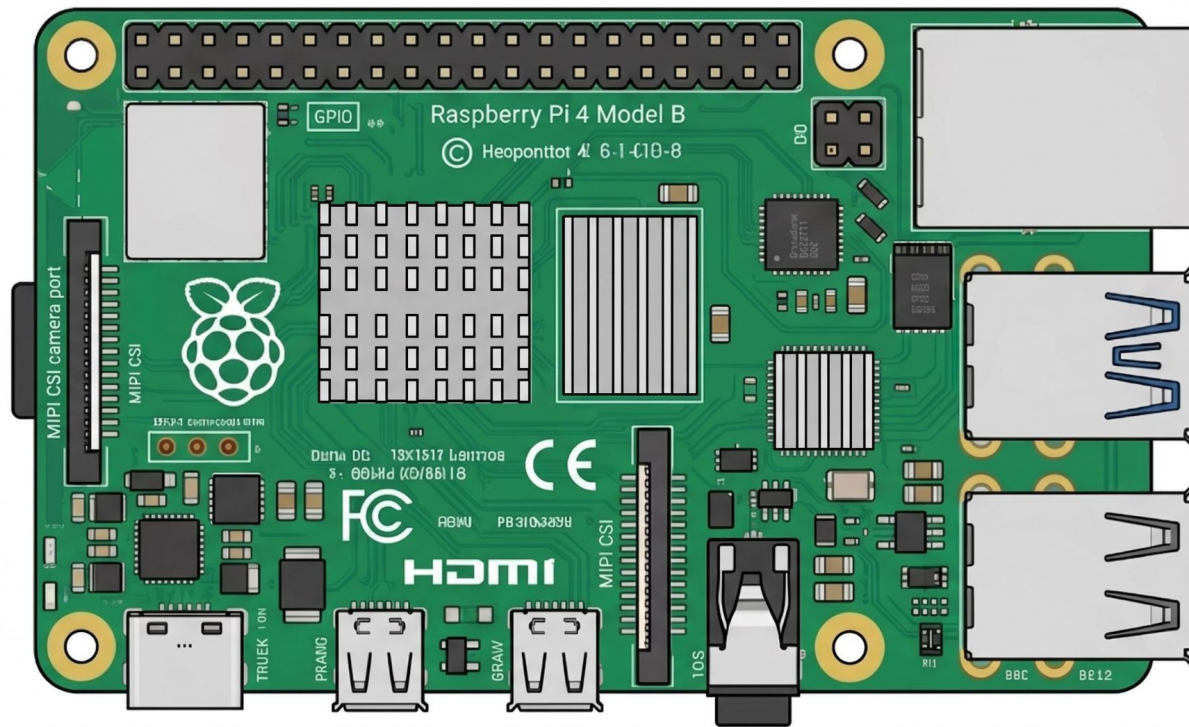
1x



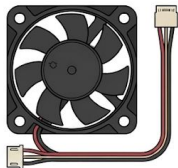
1x



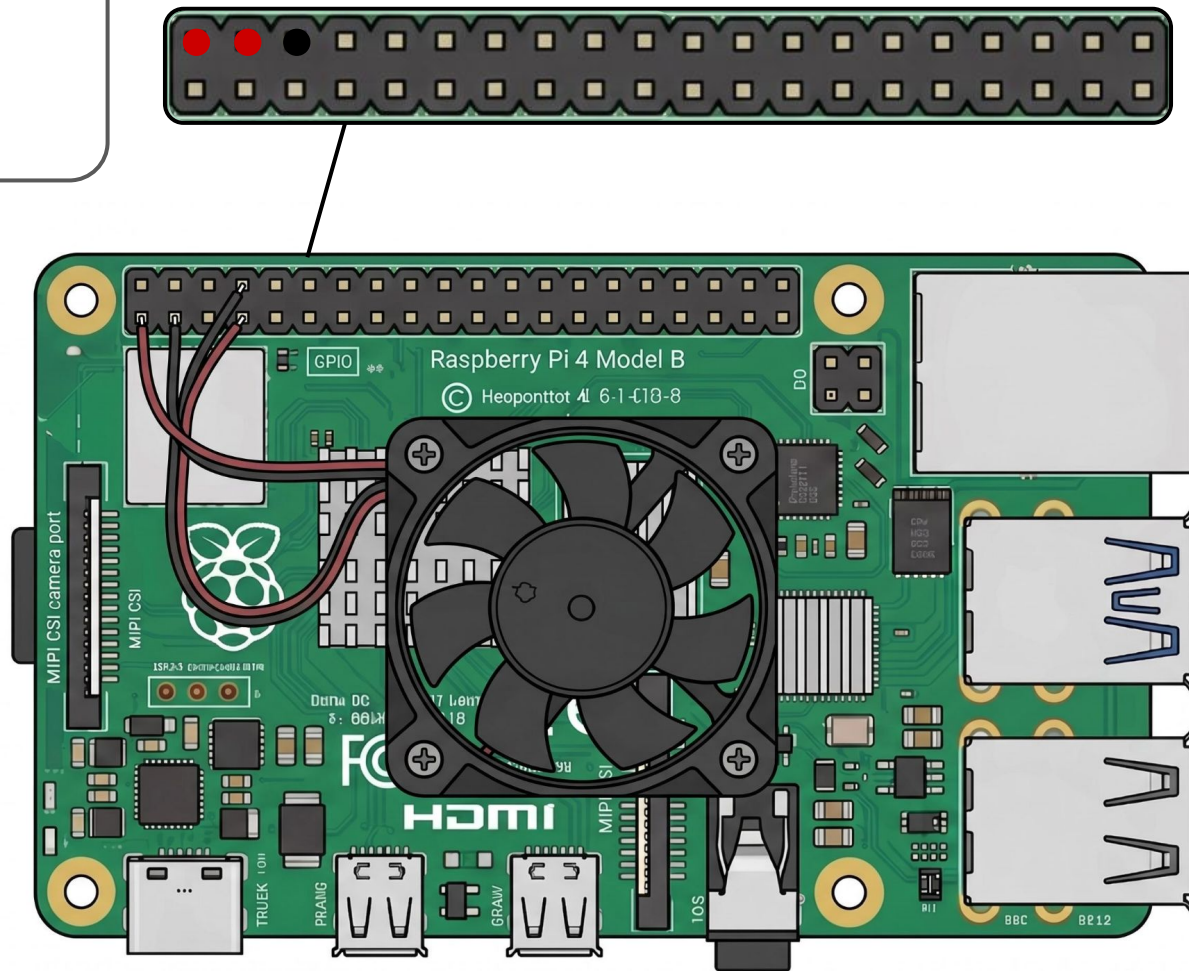
13



1x



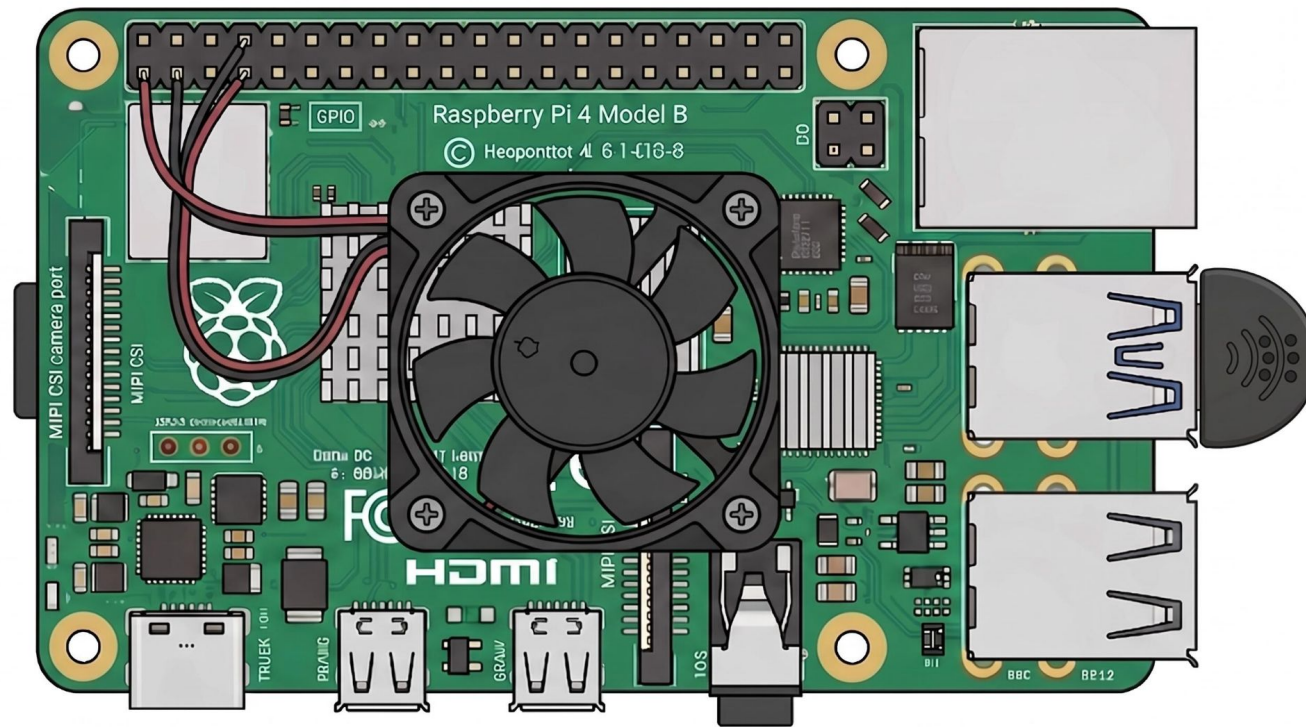
14



1x

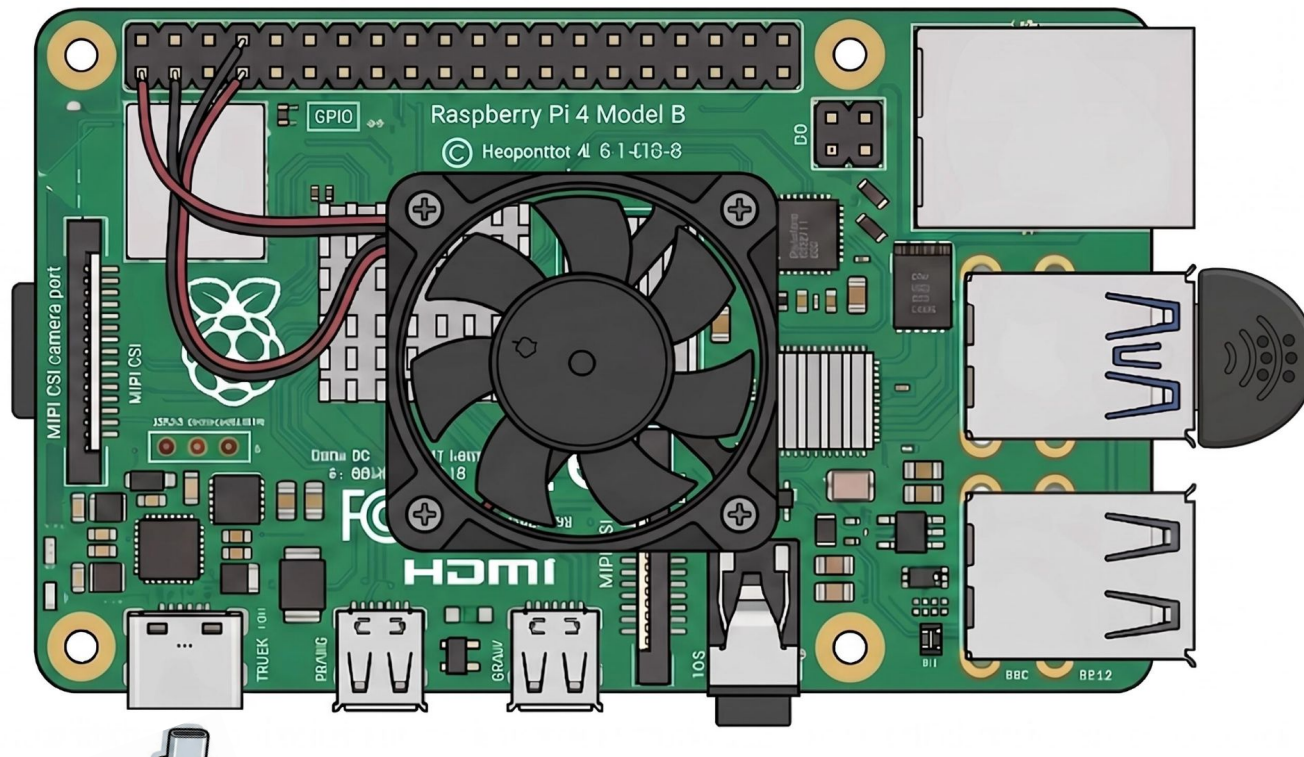


15



A black square power supply unit with the Raspberry Pi logo on top, connected to a black USB-C cable.

16



Open Terminal on Mac

Connect to your Pi → ssh
{PiName}@aslhand.local

17

```
(base) yourName@yourLaptopName ~ % ashleypi@aslhand.local
```



Open Terminal on Mac

Run these commands

18

```
ashleypi@aslhand.local:~ $ sudo apt update
ashleypi@aslhand.local:~ $ sudo apt upgrade -y
ashleypi@aslhand.local:~ $ sudo apt install -y
python3-pip i2c-tools portaudio19-dev flac
```



Open Terminal on Mac

Run this command then

“Interface Options” → “I2C” → Enable

19

```
ashleypi@aslhand.local:~ $ sudo raspi-config
```



Open Terminal on Mac

Run this command

20

```
ashleypi@aslhand.local:~ $ pip3 install  
--break-system-packages adafruit-circuitpython-pca9685  
adafruit-circuitpython-servokit SpeechRecognition pyaudio
```



Open Terminal on Mac

Test the microphone

The last line should output ~860K if the mic is working

21

```
ashleypi@aslhands.local:~ $ arecord -l
```

```
ashleypi@aslhands.local:~ $ arecord -D plughw:1,0 -d 5  
test.wav
```

```
ashleypi@aslhands.local:~ $ ls -lh test.wav
```



Part



Set Up the Hand

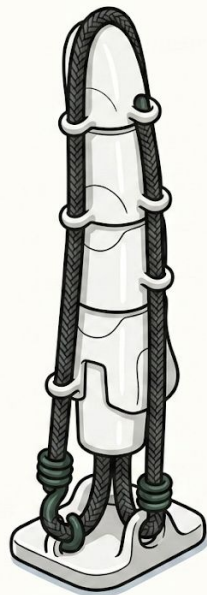
5x



1x



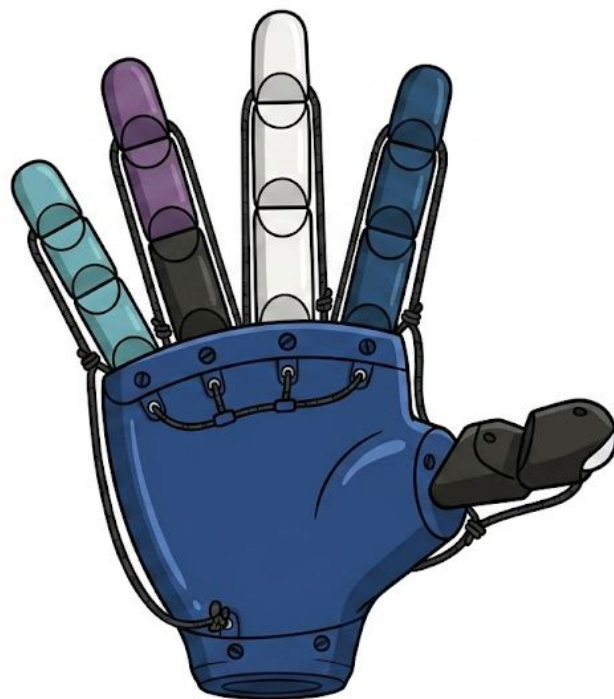
1



1x



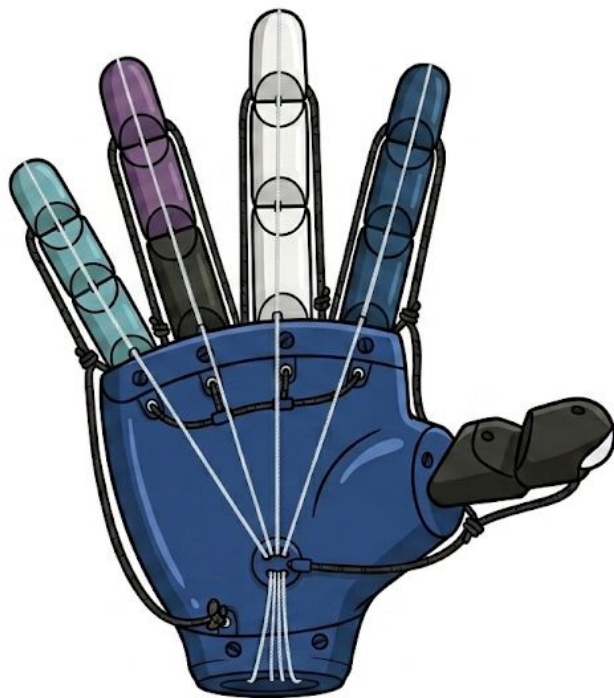
2



1x



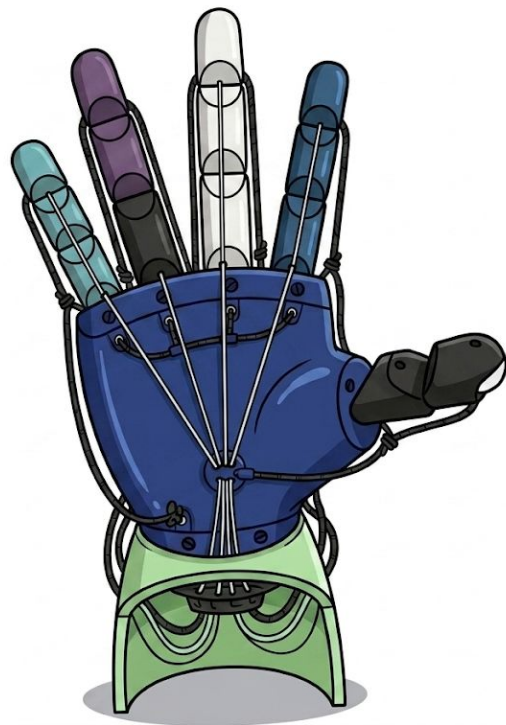
3



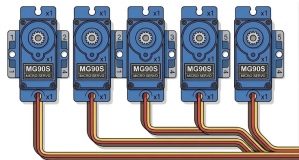
1x



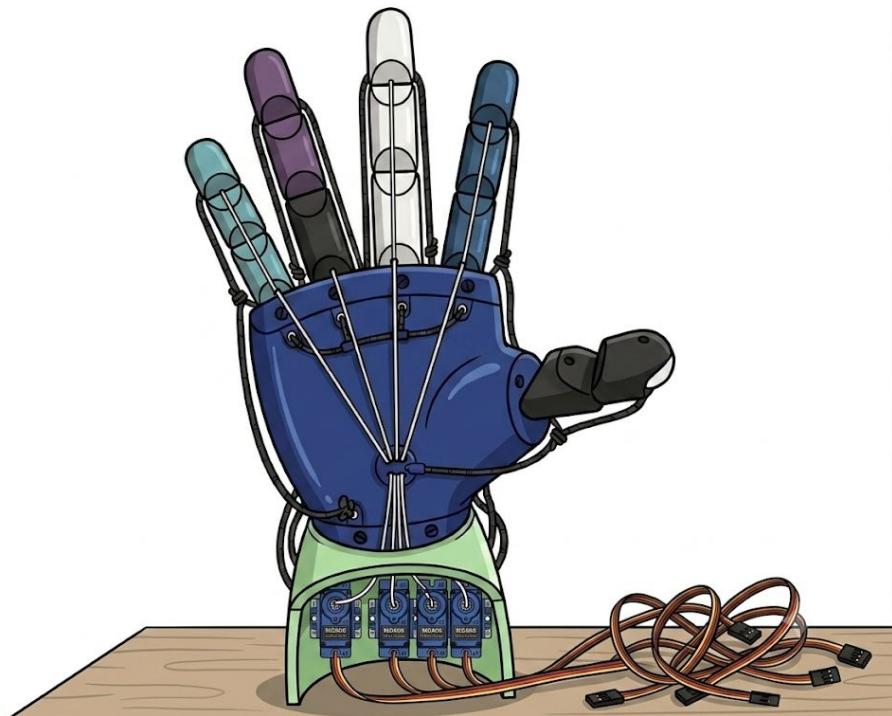
4



5x



5

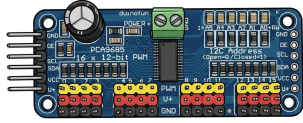


Part



Connecting the Pi to the Hand

1x



4x



1

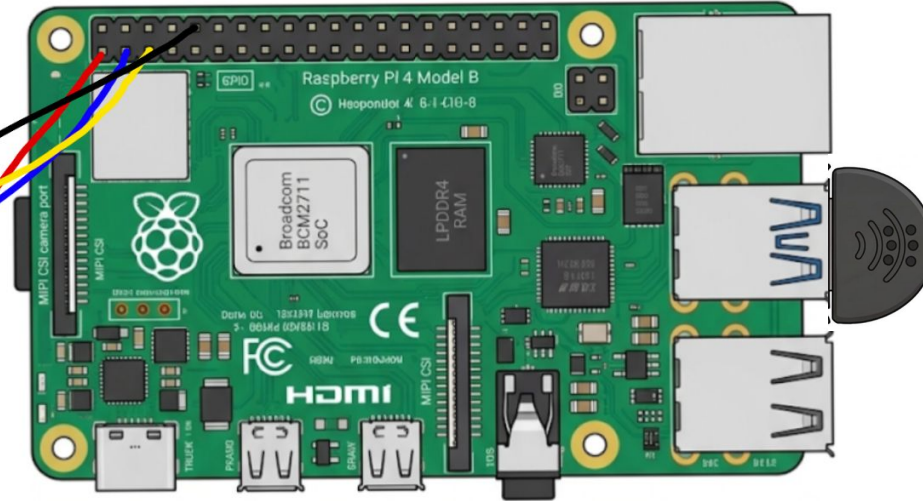
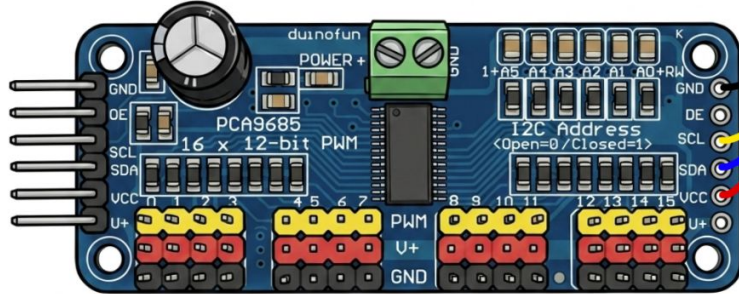
Raspberry Pi → PCA9685

Pin 1 (3.3V) → VCC

Pin 9 (GND) → GND

Pin 3 (SDA GPIO 2) → SDA

Pin 5 (SCL GPIO 3) → SCL



1x



2x

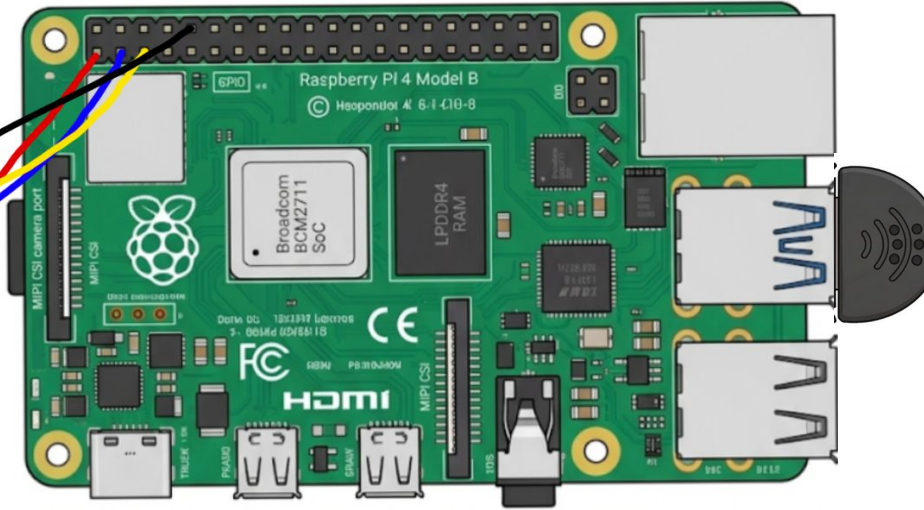
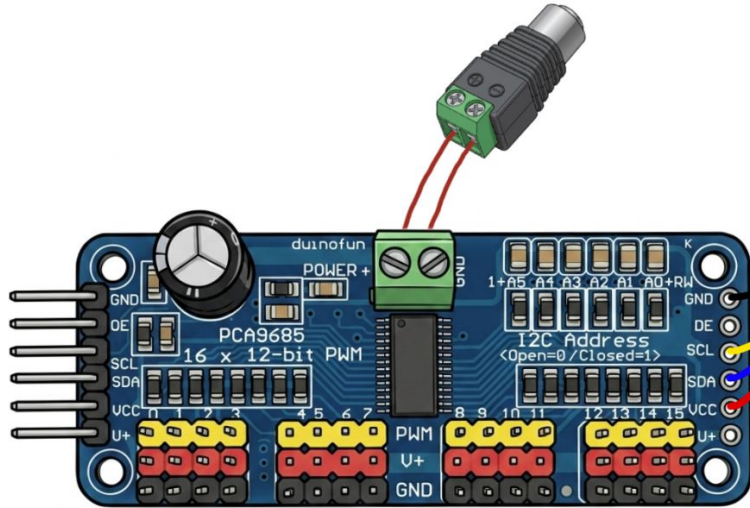


2

Barrel Jack Adapter → PCA9685

+ → V+

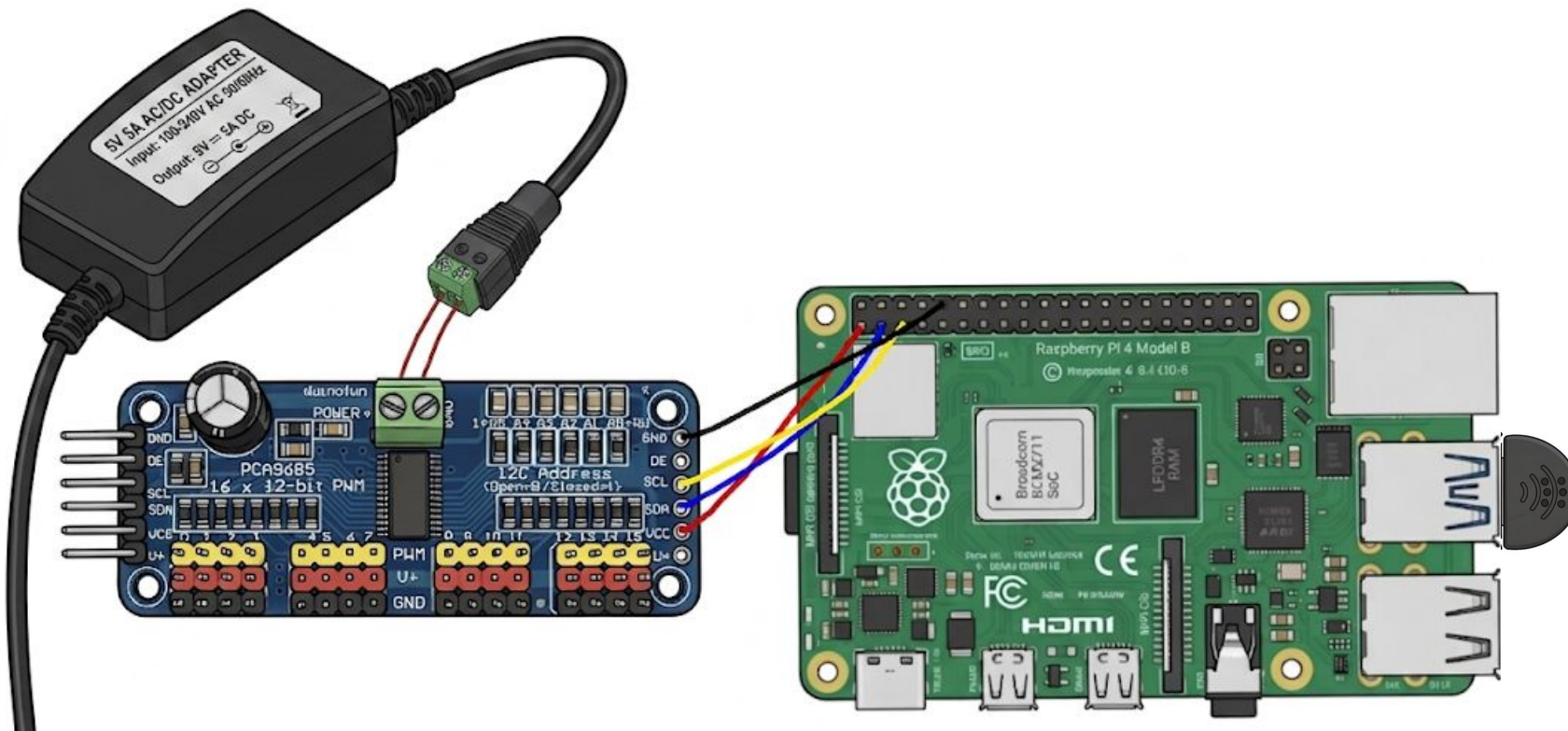
- → GND



1x



3



4

MG90S → PCA9685

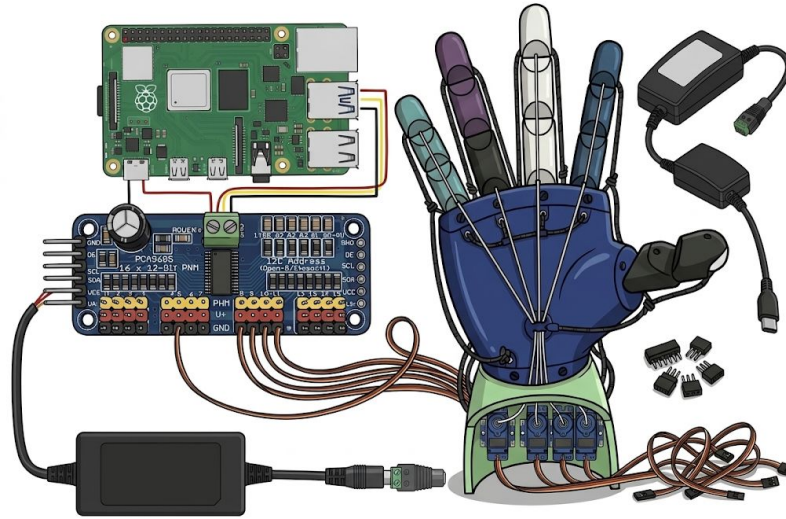
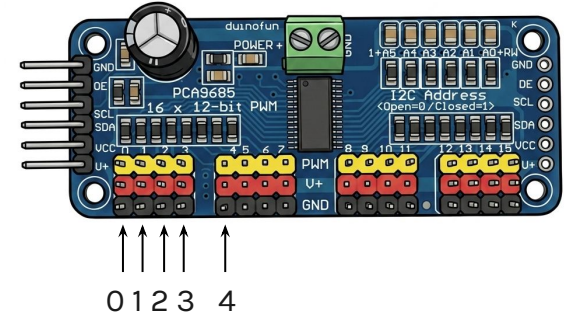
Thumb → Port 0

Index → Port 1

Middle → Port 2

Ring → Port 3

Pinky → Port 4



Make sure the yellow wire goes in the top row and the brown wire goes in the bottom row

Open Terminal on Mac

Run these commands

5

```
ashleypi@aslhand.local:~ $ nano asl_hand.py
```



Open Terminal on Mac

Paste this script then Ctrl+O to save, Enter, then Ctrl+X to exit

6

<https://tinyurl.com/5awuwn2v>



Open Terminal on Mac

Run these commands

7

```
ashleypi@as1hand.local:~ $ python3 as1_hand.py
```





YOU'RE DONE!